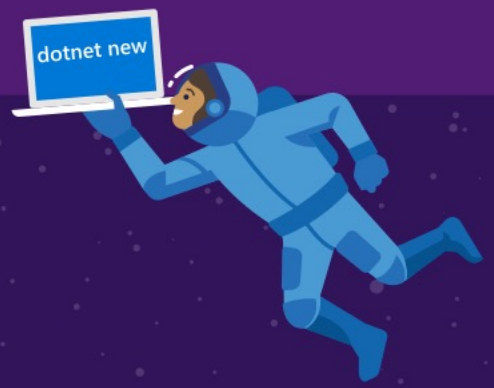
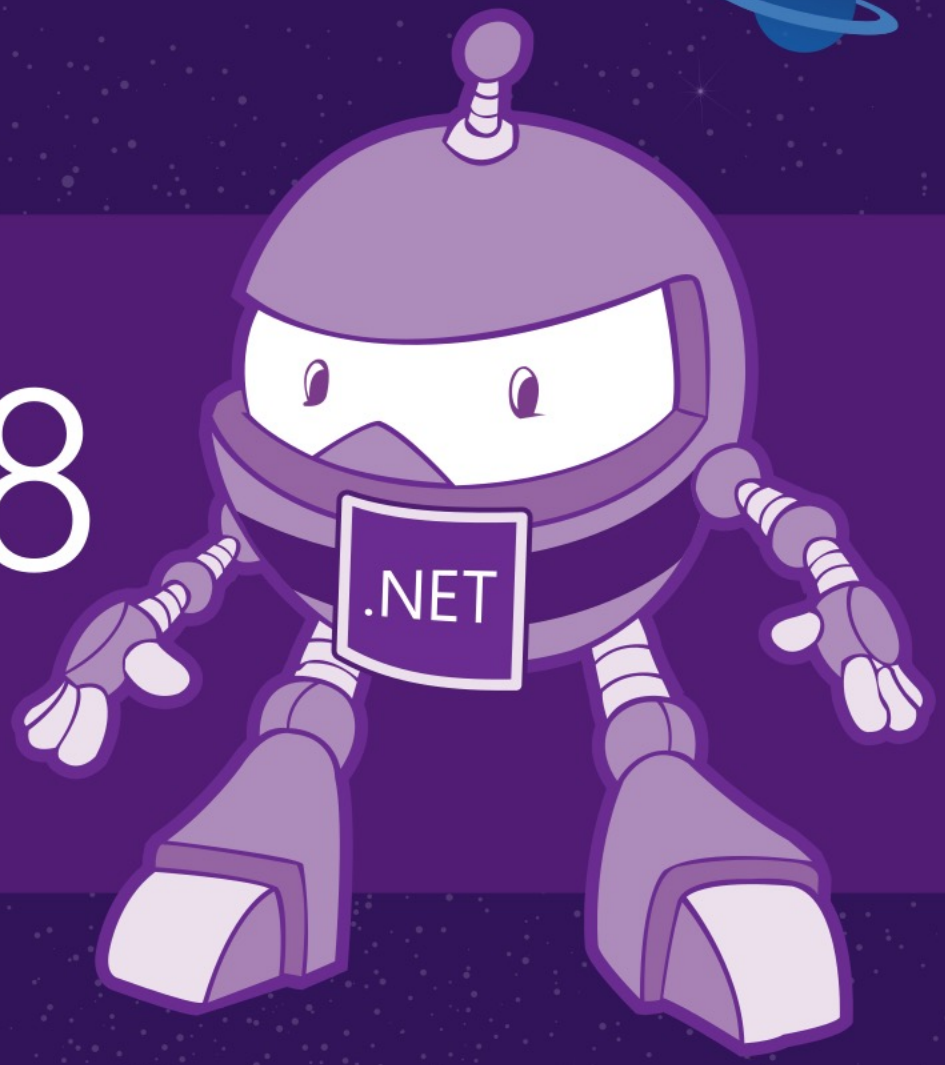
.NET Conf 2018

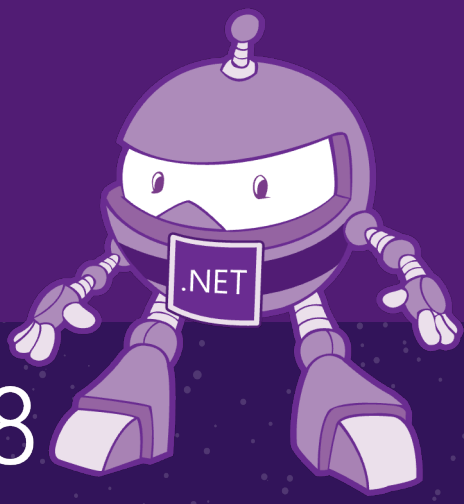
Discover the world of .NET





What's New in Xamarin.Forms

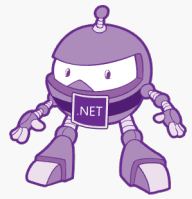
David Ortinau
Senior Program Manager, Xamarin Mobile SDKs

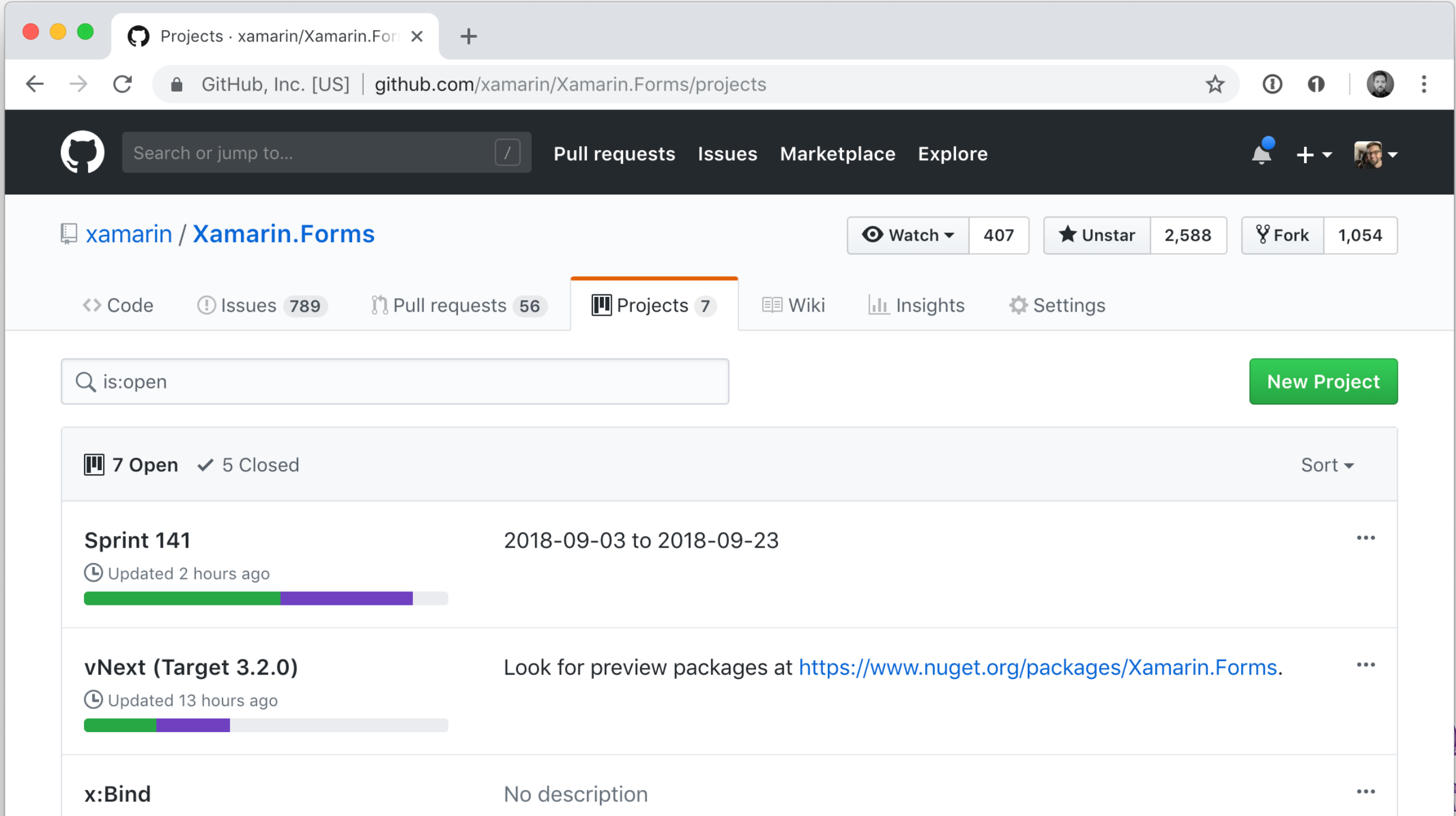


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Let's Cover Some Details

- Current Stable: 3.1.0
- Current Pre-Release: 3.2.0
- Next:
  - "Nightly" builds: https://www.myget.org/F/xamarinforms-ci/api/v2




Projects · xamarin/Xamarin.For
GitHub, Inc. [US] | github.com/xamarin/Xamarin.Forms/projects
Search or jump to...
Pull requests
Issues
Marketplace
Explore
xamarin / Xamarin.Forms
Watch 407
Unstar 2,588
Fork 1,054
Code
Issues 789
Pull requests 56
Projects 7
Wiki
Insights
Settings
is:open
New Project
7 Open
5 Closed
Sort
Sprint 141
Updated 2 hours ago
2018-09-03 to 2018-09-23
vNext (Target 3.2.0)
Updated 13 hours ago
Look for preview packages at https://www.nuget.org/packages/Xamarin.Forms.
x:Bind
No description

Feature Roadmap

Edit New Page

David Ortinau edited this page 11 hours ago · 25 revisions

This roadmap outlines our anticipated major feature releases in addition to proposed and planned ongoing service releases.

> **Disclaimer** We cannot predict the future and how everything will shake out. Things will change. Timing may be adjusted due to priority changes, in pursuit of quality standards, or any number of other really good reasons that we will strive to proactively and openly communicate.

Current Version

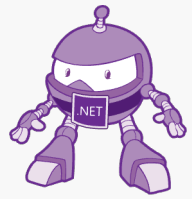
Pages 8

General Info

- Feature Roadmap
- How to Build Xamarin.Forms NuGet
- How to Contribute
- Nightly Builds
- Submitting Issues

Let's Talk About

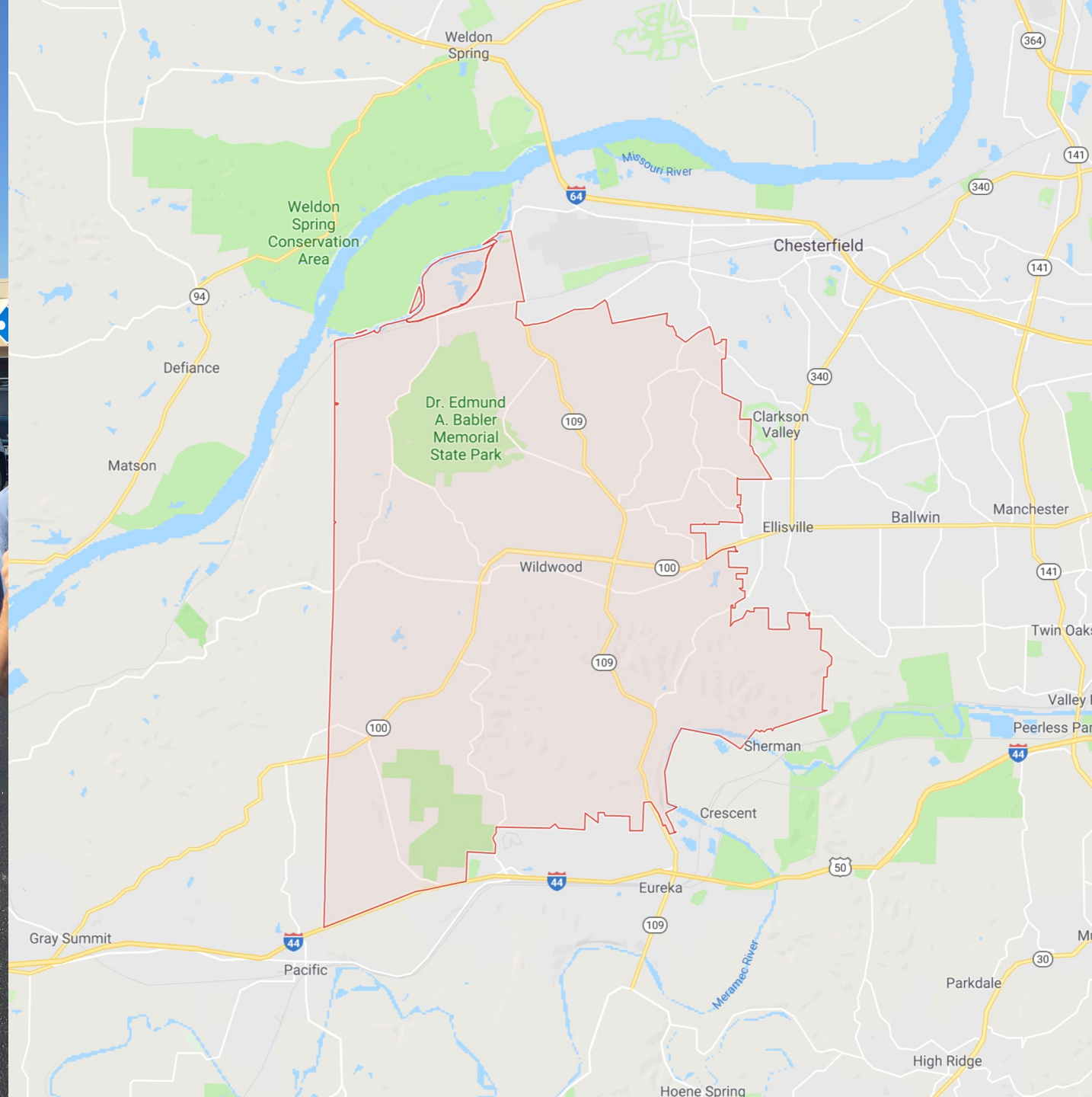
- Big Things
- Little Things
- Next Things
  - Yes, I'm making you stick around for this one. ☺



Big Things




Stanford
DENTAL
STOP


Weldon Spring
Missouri River
Weldon Spring Conservation Area
Chesterfield
Defiance
Dr. Edmund A. Babler Memorial State Park
Clarkson Valley
Matson
Ellisville
Ballwin
Manchester
Wildwood
Twin Oaks
Sherman
Peerless Park
Crescent
Eureka
Gray Summit
Pacific
Meramec River
Parkdale
High Ridge
Hoene Spring
64
94
109
340
141
100
44
50
30
364

XAMLC and Compiled Bindings

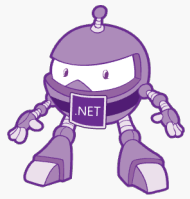
Perf

```
[assembly: XamlCompilation (XamlCompilationOptions.Compile)]
Namespace MyApp { ....

<ListView x:Name="colorListView"
                Grid.Row="0"
                ItemsSource="{x:Static local:NamedColor.All}"
                RowHeight="40">
          <ListView.ItemTemplate>
             <DataTemplate x:DataType="{x:Type local:NamedColor}">
                <ViewCell>
                   <StackLayout Orientation="Horizontal">
                      <BoxView Color="{Binding Color}"
                               HeightRequest="32"
                               WidthRequest="32"
                               VerticalOptions="Center" />
```

…

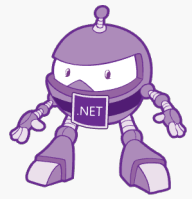
https://developer.xamarin.com/guides/xamarin-forms/application-fundamentals/data-binding/compiled-bindings/



Android Fast Scrolling

Perf

```
<ContentPage
     xmlns="http://xamarin.com/schemas/2014/forms"
     xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
     x:Class="TheLittleThingsPlayground.Views.ItemsPage"
     Title="{Binding Title}" x:Name="BrowseItemsPage"
     xmlns:android="clr-namespace:Xamarin.Forms.PlatformConfiguration.AndroidSpecific;assembly=Xamarin.Forms.Core">
    <ContentPage.Content>
        <StackLayout>
            <ListView x:Name="ItemsListView"
                      android:ListView.IsFastScrollEnabled="true"
…
</ContentPage>
```

Android Fast Renderers

Perf

Button, Image, Label

<Button />

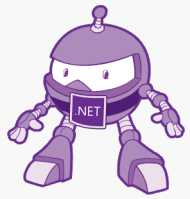
Default

ButtonRenderer

AppCompatButton

Fast Renderer

AppCompatButton

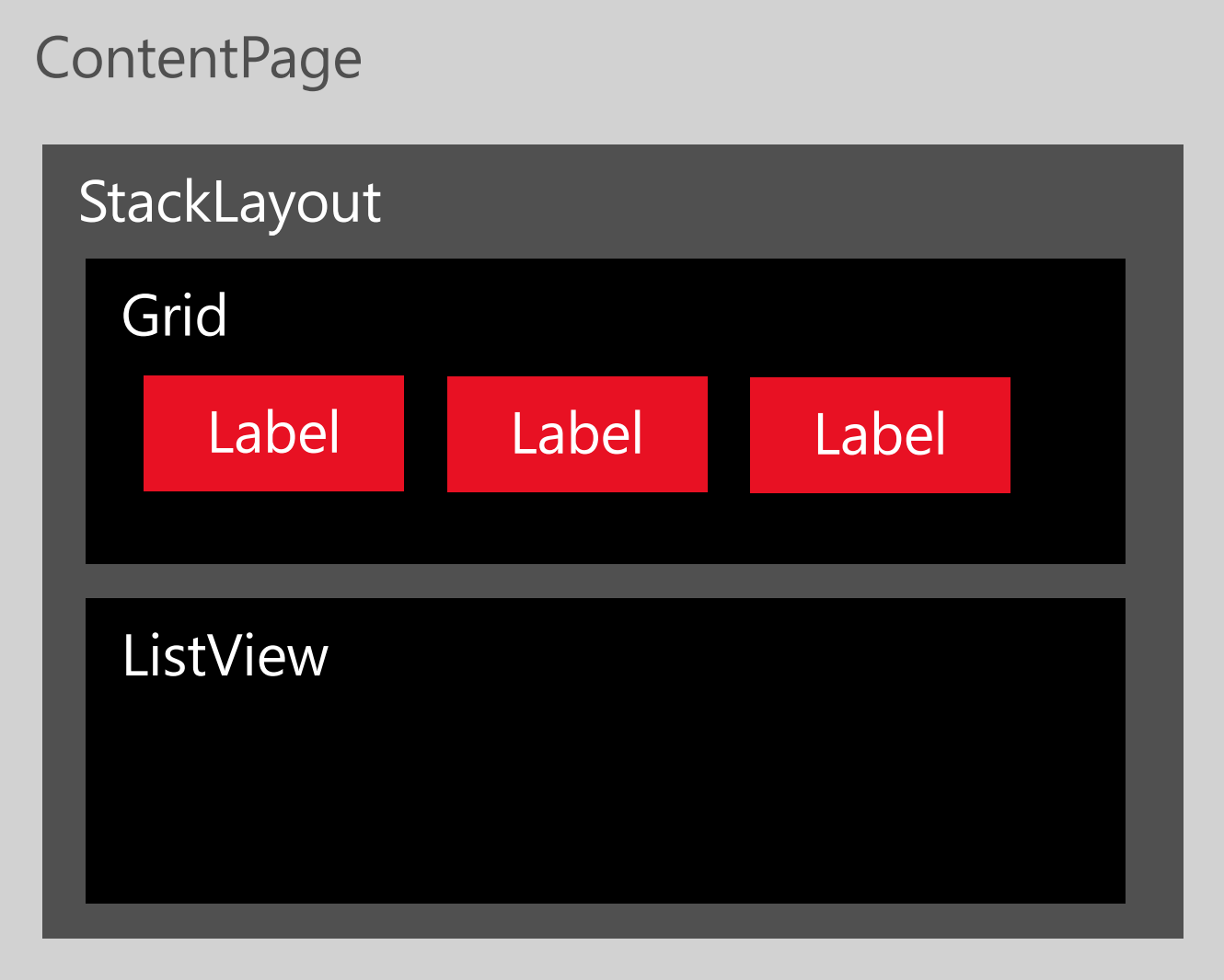


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```
Forms.SetFlags("FastRenderers_Experimental");
Forms.Init(this, bundle);
```

Layout Compression

Perf




```
<StackLayout
             Spacing="0"
             CompressedLayout.IsHeadless="true"
             VerticalOptions="FillAndExpand"
             HorizontalOptions="FillAndExpand">
    <Grid
         CompressedLayout.IsHeadless="true”>
    ...
    </Grid>
...
</StackLayout>
```



Uncompressed
130 Views

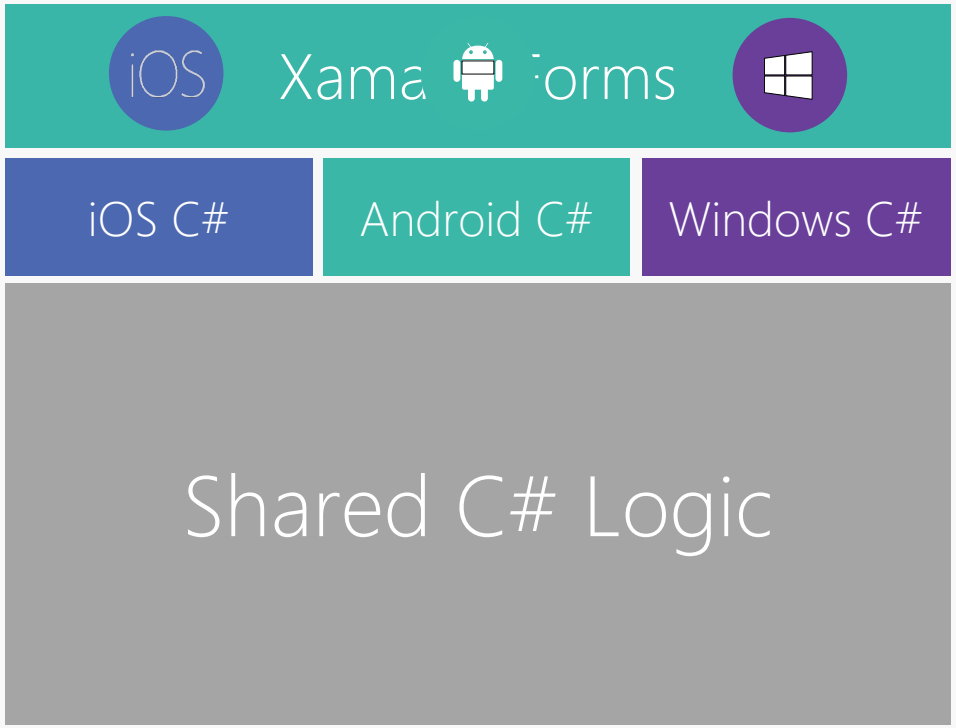


Compressed
111 Views

Compressed + Fast Renderers
70 Views

Native Forms

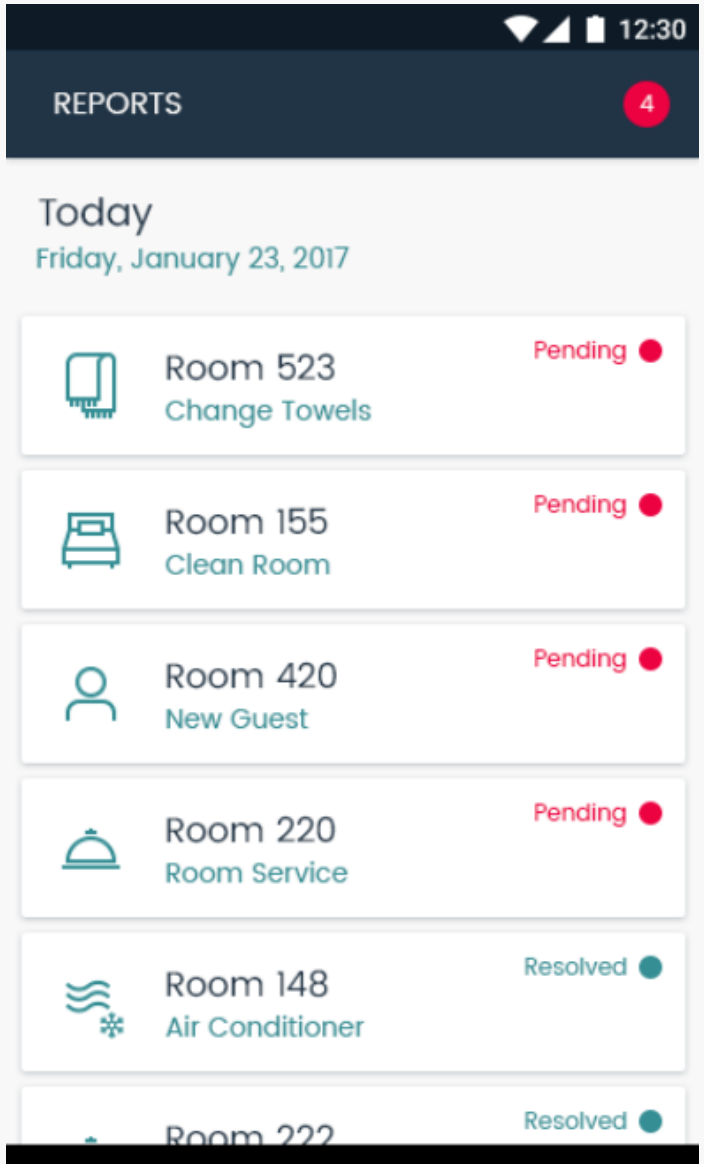
Perf



Xamarin.Forms Supported Features

- XAML Compilation
- Bindings
- MessagingCenter
- DependencyService
- Create
  - UIViewController
  - Activity
  - Fragment
  - FrameworkElement

SmartHotel360: Native Forms

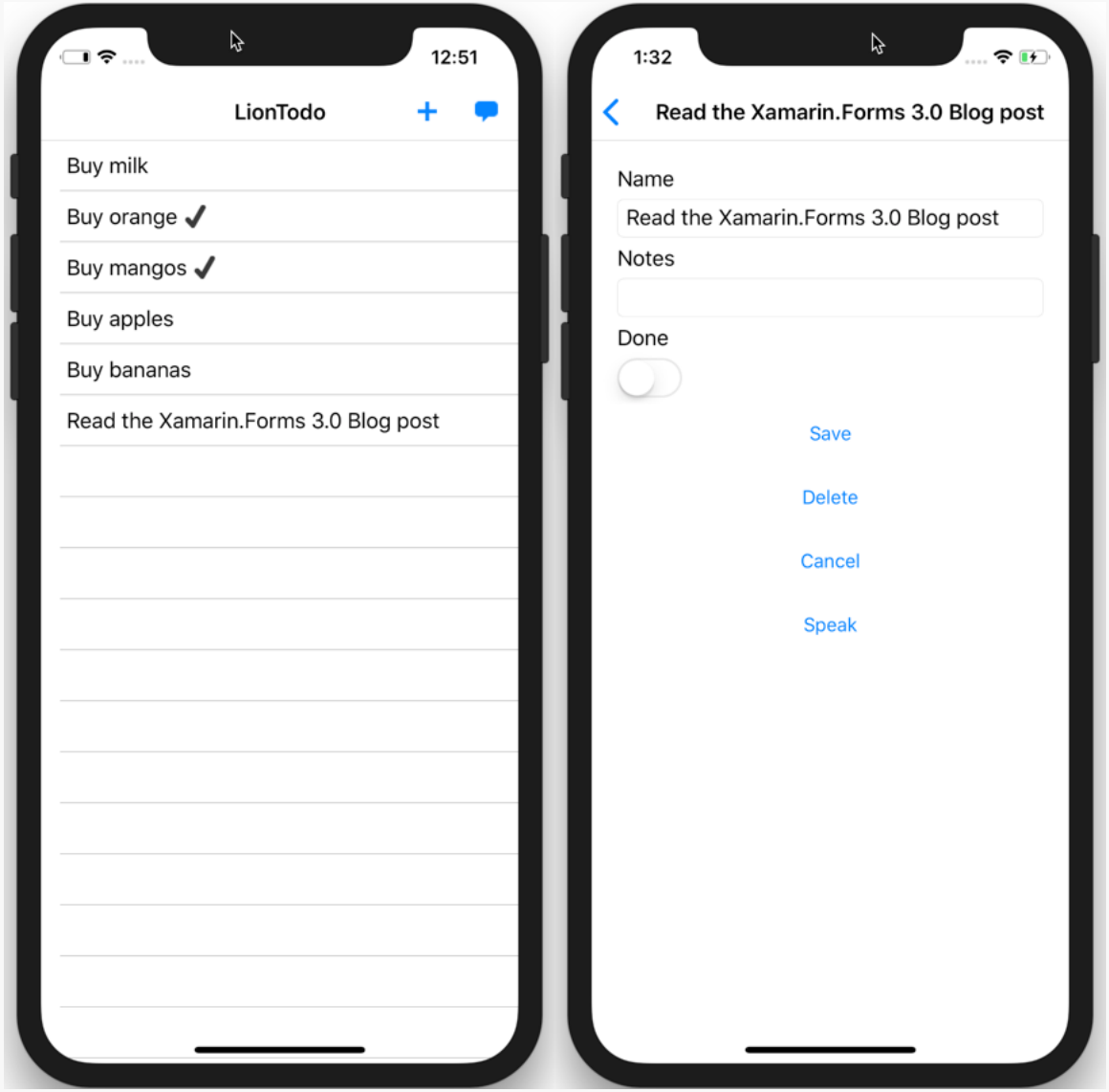



```
// Minimal setup 1x before instantiating a ContentPage
Xamarin.Forms.Forms.Init();

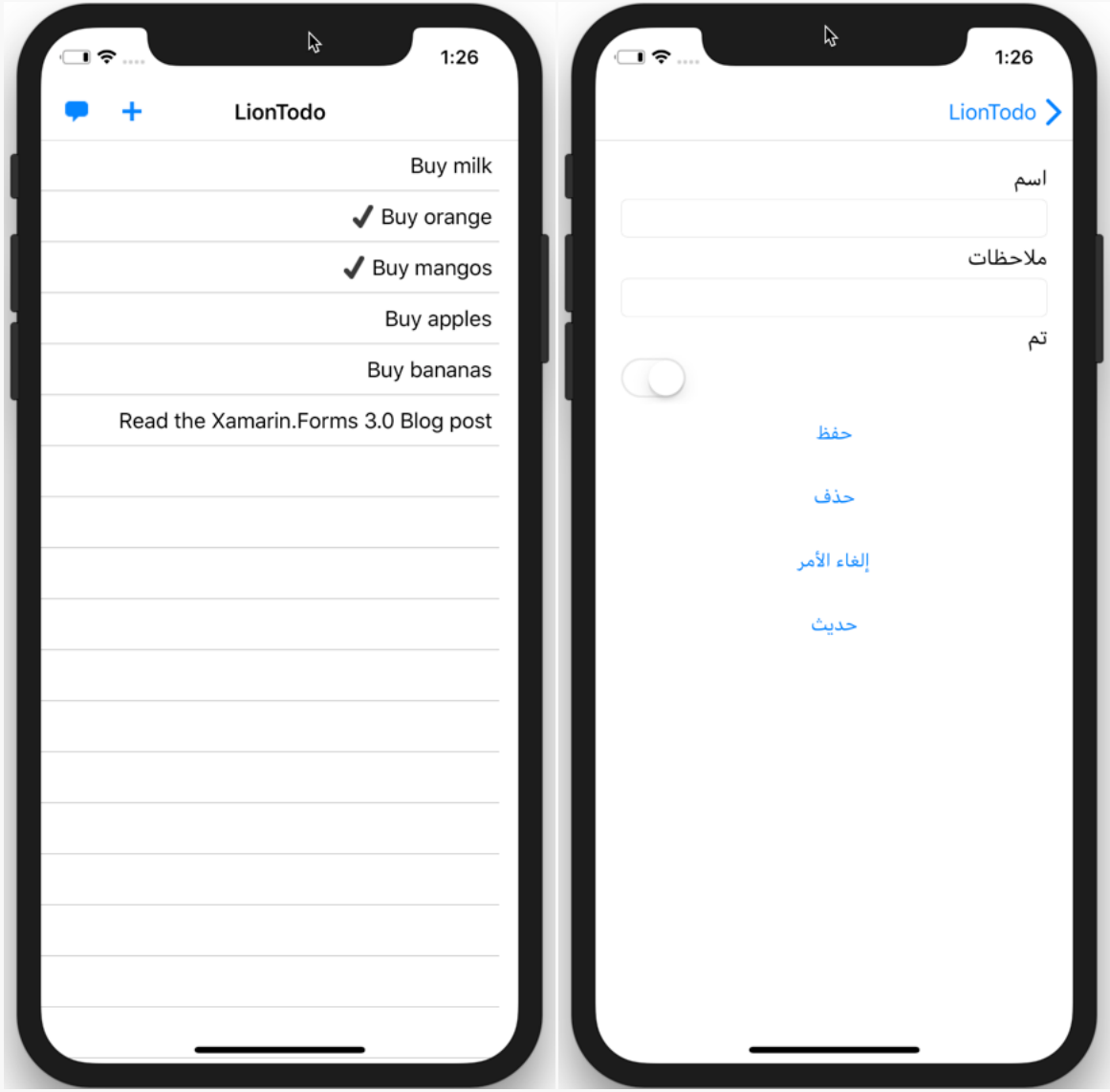
var settingsView = new SettingsView
{
    BindingContext = settingsViewModel
};

var settingsController = settingsView.CreateViewController();
NavigationController.PushViewController(settingsController, true);
```

LTR


12:51
LionTodo
Buy milk
Buy orange ✓
Buy mangos ✓
Buy apples
Buy bananas
Read the Xamarin.Forms 3.0 Blog post
1:32
Read the Xamarin.Forms 3.0 Blog post
Name
Read the Xamarin.Forms 3.0 Blog post
Notes
Done
Save
Delete
Cancel
Speak

RTL

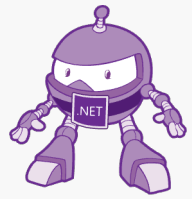

1:26
LionTodo
Buy milk
✓ Buy orange
✓ Buy mangos
Buy apples
Buy bananas
Read the Xamarin.Forms 3.0 Blog post
1:26
LionTodo
اسم
ملاحظات
تم
حفظ
حذف
إلغاء الأمر
حديث

FlowDirection="{x:Static Device.FlowDirection}"

FlexLayout

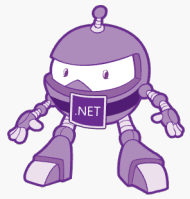
- Flexbox inspired layout system
- Prioritizes distribution and alignment
- Great for adaptive layouts – multiple form factors
- Promotes flatter layout, reducing UI nesting

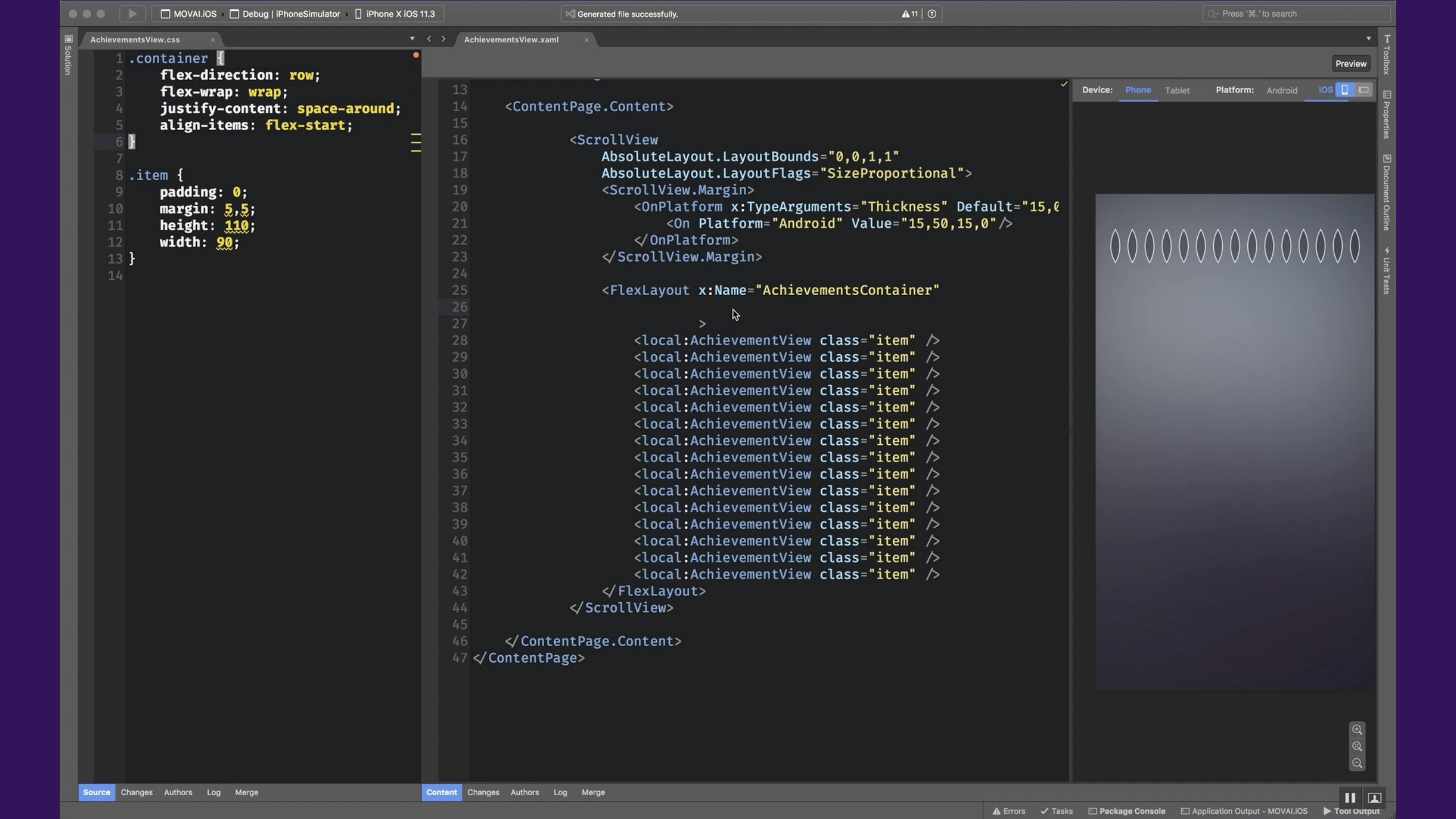
https://blog.xamarin.com/adaptive-mobile-designs-with-flexlayout/



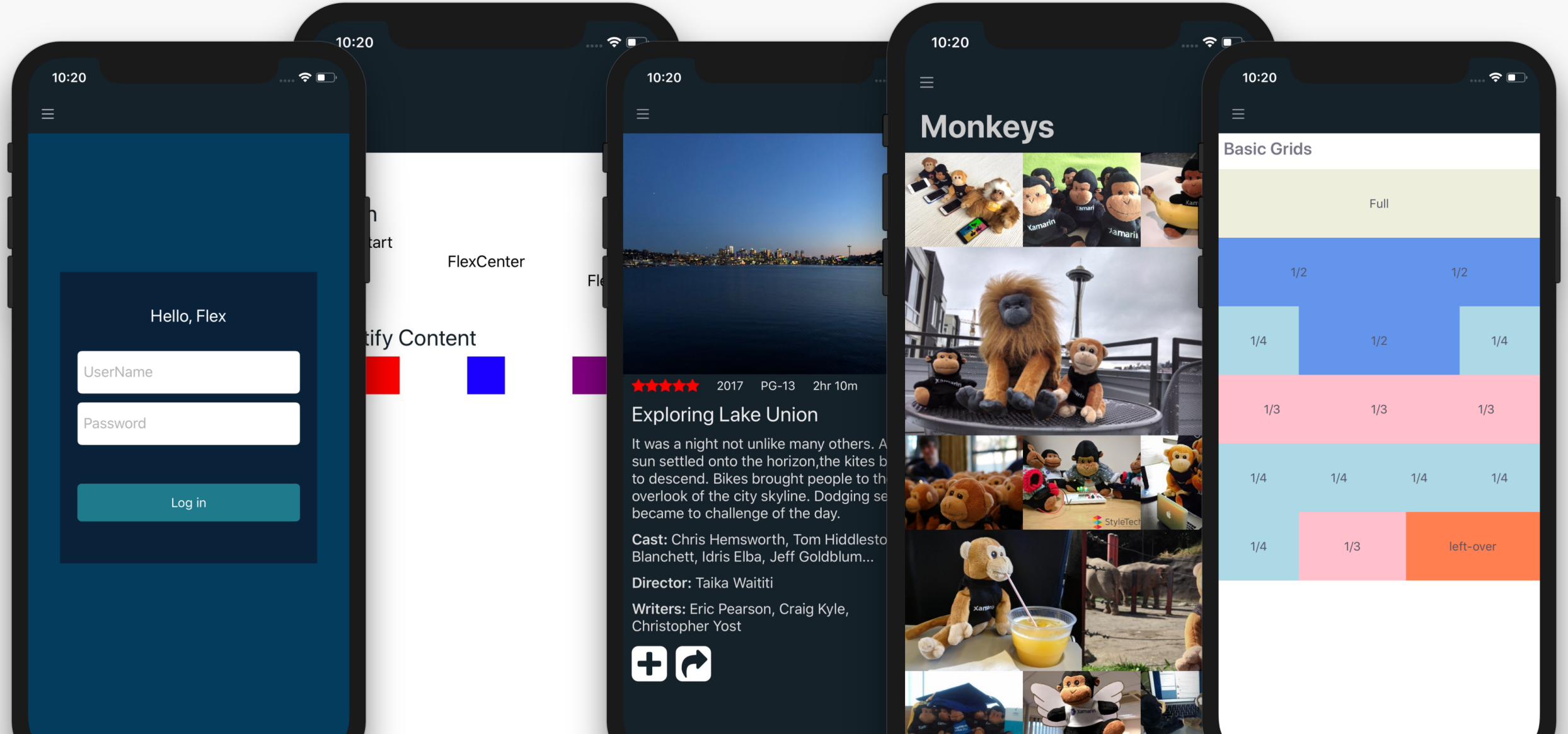
CSS

- Styles expressed via Cascading Style Sheets (CSS)
- Parsed a runtime
- Does not replace XAML Styles
- Supports many properties
- But not all…
- Styles are applied one by one, in definition order
- Styles defined on the item itself is always applied last




MOVAI.iOS › Debug | iPhoneSimulator › iPhone X iOS 11.3
Generated file successfully.
Press '⌘.' to search
Solution
AchievementsView.css
.container {
flex-direction: row;
flex-wrap: wrap;
justify-content: space-around;
align-items: flex-start;
}
.item {
padding: 0;
margin: 5,5;
height: 110;
width: 90;
}
AchievementsView.xaml
Preview
Device: Phone Tablet Platform: Android iOS
<ContentPage.Content>
<ScrollView
AbsoluteLayout.LayoutBounds="0,0,1,1"
AbsoluteLayout.LayoutFlags="SizeProportional">
<ScrollView.Margin>
<OnPlatform x:TypeArguments="Thickness" Default="15,0
<On Platform="Android" Value="15,50,15,0"/>
</OnPlatform>
</ScrollView.Margin>
<FlexLayout x:Name="AchievementsContainer"
>
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
</FlexLayout>
</ScrollView>
</ContentPage.Content>
</ContentPage>
Source Changes Authors Log Merge
Content Changes Authors Log Merge
Toolbox
Properties
Document Outline
Unit Tests
Errors Tasks Package Console Application Output - MOVAI.iOS Tool Output

http://github.com/davidortinau/Flexibility



Visual State Manager

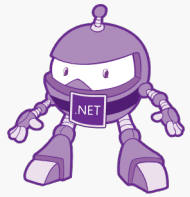
- VisualElement "CommonStates"
  - Normal
  - Focused
  - Disabled

- GoToState(this, "Focused");

```
<Entry FontSize="18">
    <VisualStateManager.VisualStateGroups>
        <VisualStateGroup x:Name="CommonStates">
            <VisualState x:Name="Normal">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Lime" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Focused">
                <VisualState.Setters>
                    <Setter Property="FontSize"
                            Value="36" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Disabled">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Pink" />
                </VisualState.Setters>
            </VisualState>
        </VisualStateGroup>
    </VisualStateManager.VisualStateGroups>
</Entry>
```



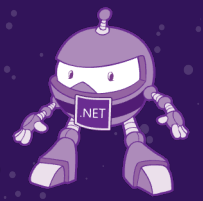
http://github.com/microsoft/ConferenceVision



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Little Things





F100: Addressing the Little Things

[Project Board]

Entry/Editor

- IsSpellCheckEnabled
- MaxLength
- DetectReadingOrderFromContent (UWP)
- Text prediction options
- Autocapitalization
- AutoResizable Editor
- ButtonType for return button
- Position and Color of caret
- Placeholder color

Label

- Bindable Spans
- Commandable Spans, Span Gestures
- LineHeight

ListView

- full width separators
- SelectionMode

SearchBar IsSpellCheckEnabled

ImeOptions

DesignModeEnabled

Android Bottom Tabs

WebView mixed content (Android)

WebView evaluate javascript

Binding Mode OneTime

ProgressBar colors

Switch colors

Slider colors

Hide scroll bars

Button AccessKey

TabbedPage Icons (UWP)

Drop Shadows (iOS/UWP)

App DI Container Hooks

Button Padding

Page TitleView

BoxView rounded corners

WKWebView for iOS

Coming Soon

Label MaxLines

Entry Read-Only

ImageButton

GIF Support

Button and Label Parity

PinsSource on Forms.Map

WebView Zoom Controls (Android)

Help Wanted

Rounded Corners

Rounded Image

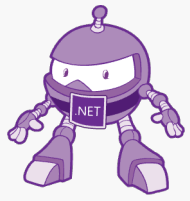
ProgressBar.Height

InkWell Ripple Effect

Size WebView to Content

Floating Action Button

Android Theme Capitalization override

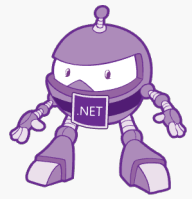



IN RECOGNITION OF YOUR
XAMARIN.FORMS CONTRIBUTIONS
Almir Vuk
Daniel Hindrikes
June, 2018
HACK DAY
July 2018
XAMARIN CERTIFIED
MOBILE DEVELOPER
MARTIN ZIKMUND
Felipe Momm

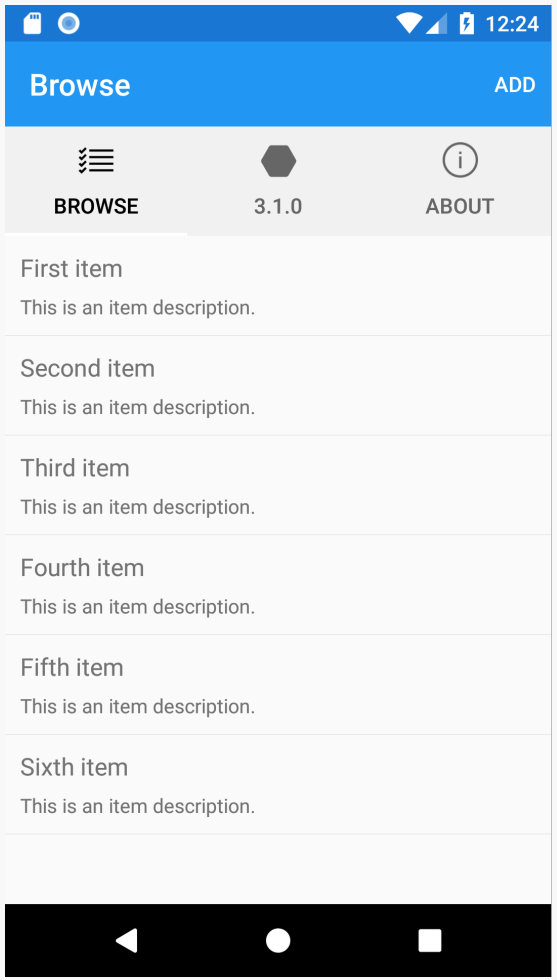
THANK YOU!

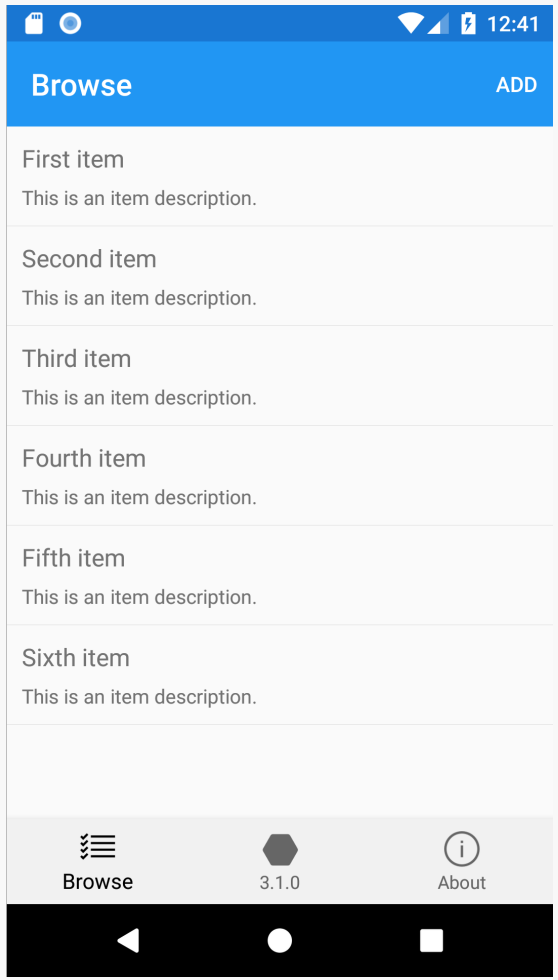
Join the Fun!

- https://blog.xamarin.com/beginners-guide-contributing-xamarin-forms/
- https://blog.xamarin.com/debugging-xamarin-forms/



Android Bottom Tabs

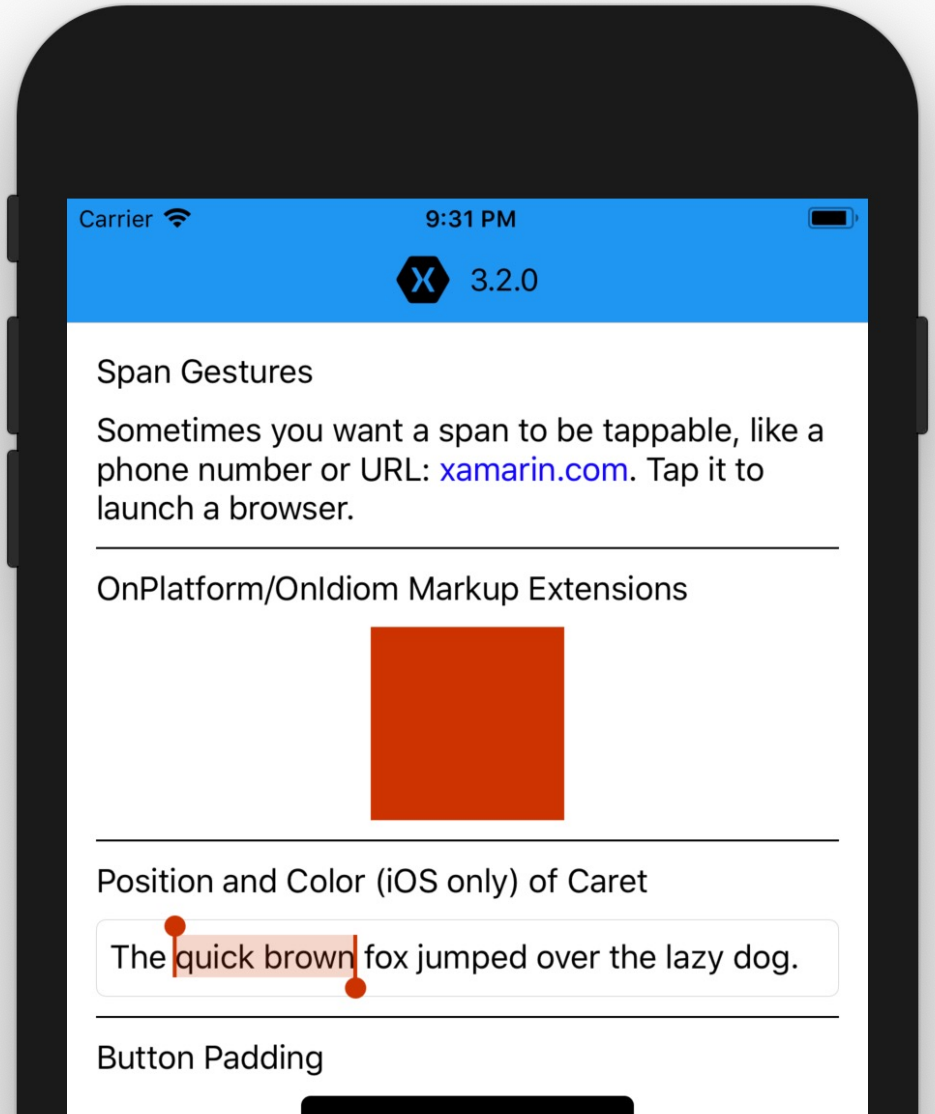





```
<TabbedPage …
            BarBackgroundColor="#F1F1F1"
            xmlns:android="clr-
namespace:Xamarin.Forms.PlatformConfiguration.AndroidSpecific;assembly
=Xamarin.Forms.Core"
            android:TabbedPage.ToolbarPlacement="Bottom"
            android:TabbedPage.BarItemColor="#666666"
            android:TabbedPage.BarSelectedItemColor="Black">

```

TitleView

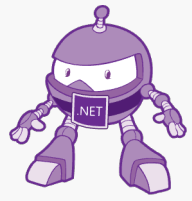



```
<ContentPage>
    <NavigationPage.TitleView>
        <StackLayout
            Orientation="Horizontal"
            VerticalOptions="Center" Spacing="10">
            <Image Source="iconXamagon.png">
                <Image.GestureRecognizers>
                    <TapGestureRecognizer
                        Tapped="Handle_Tapped"/>
                </Image.GestureRecognizers>
            </Image>
            <Label Text="3.2.0"
                FontSize="16"
                TextColor="Black"
                VerticalTextAlignment="Center" />
        </StackLayout>
    </NavigationPage.TitleView>
    ...
</ContentPage>
```

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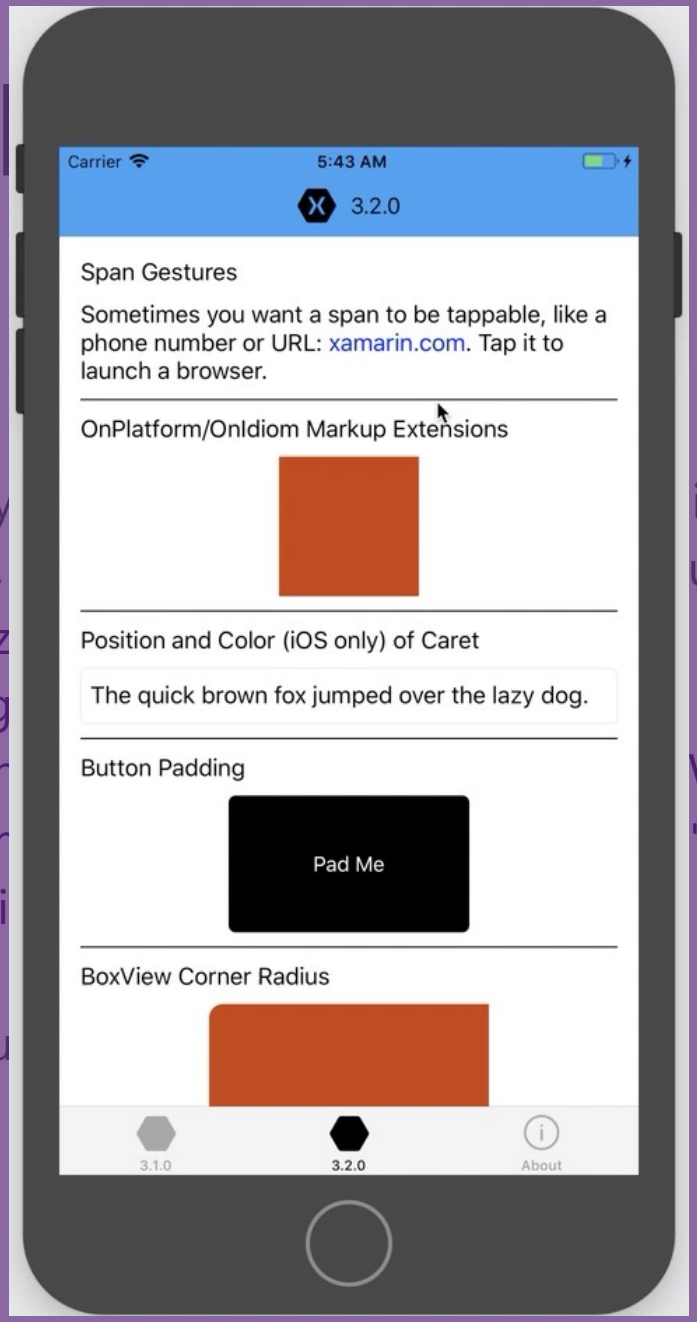
Bindable Span

```
<Label>
    <Label.FormattedText>
        <FormattedString>
            <Span Text="Welcome " />
            <Span Text="{Binding YourName}" FontAttributes="Bold"/>
            <Span Text=" to the Xamarin.Forms 3.1.0 playground!" />
        </FormattedString>
    </Label.FormattedText>
</Label>
```

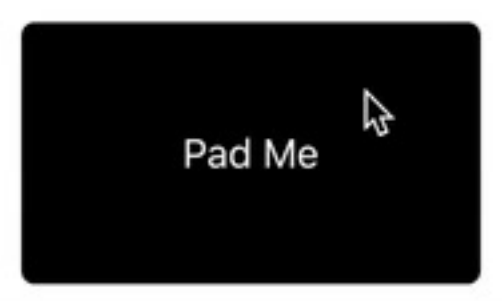
Commandable Sp

```
<Label>
    <Label.FormattedText>
      <FormattedString>
        <Span Text="Sometimes y                                    ike a phone number or URL: " />
        <Span Text="{Binding Url,                                  ue">
          <Span.GestureRecogniz
            <TapGestureRecog
              Command="{Bir                                        Way}"
              CommandParam                                         "/>
          </Span.GestureRecogni
        </Span>
        <Span Text=". Tap it to lau
      </FormattedString>
    </Label.FormattedText>
</Label>
```



Button Padding

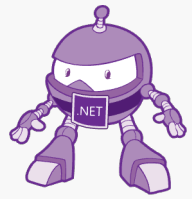
```
<Button Text="Pad Me"
        Padding="60,40"
        BackgroundColor="Black"
        TextColor="White"/>
```



OnPlatform & OnIdiom XAML Extensions

- Concise syntax
- Supports Default value
- Support Converter and ConverterParamter

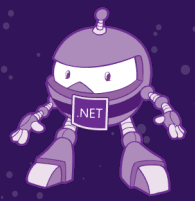
```
<BoxView
    FlexLayout.AlignSelf="Center"
    WidthRequest="100"
    HeightRequest="100"
    BackgroundColor="{OnPlatform Android=#333333, iOS=#CC3300, Default=Green}" />
```



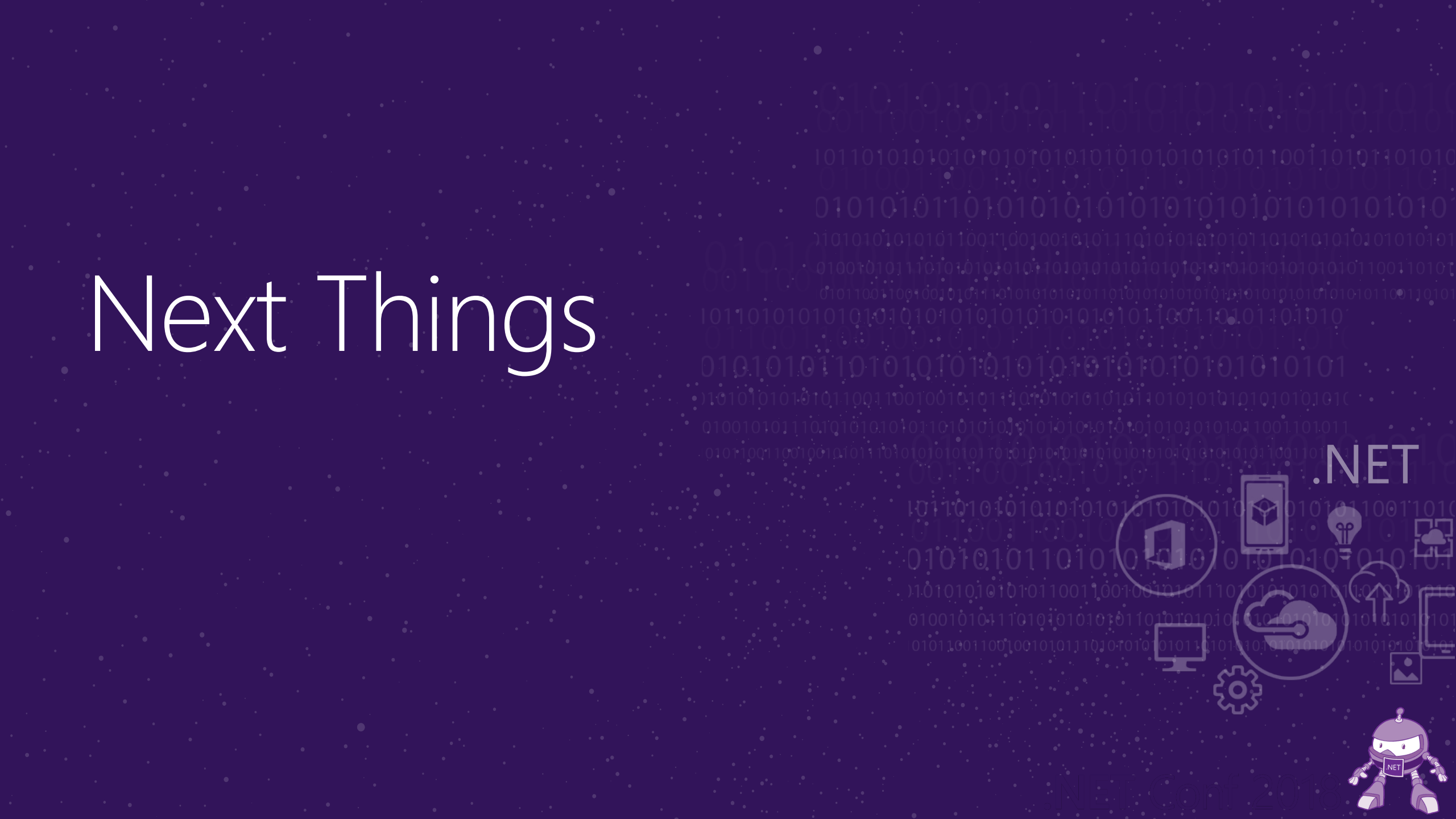
https://github.com/davidortinau/TheLittleThingsPlayground

.NET





Next Things



Xamarin Essentials

A kit of essential cross platform APIs for your mobile apps – Native Android, Native iOS or Forms



Accelerometer

App Information

Battery State

Clipboard

Compass

Connectivity

Data Transfer

Device Information

File System Helpers

Email

Geocoding

Geolocation

Gyroscope

Magnetometer

Open Browser

Phone Dialer

Preferences

Screen lock

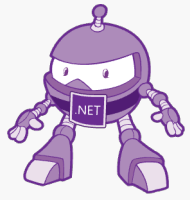
Secure Storage

SMS

Text to Speech

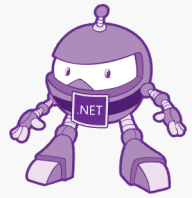
Version tracking

Vibrate



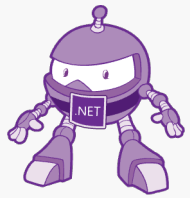
More CSS Properties

- ActivityIndicator, BoxView, ProgressBar, Switch
  - color
- Grid
  - row-gap
  - column-gap
- VisualElement
  - transform
  - transform-origin
- Label
  - vertical-align



New Xamarin.Forms Specific CSS Properties

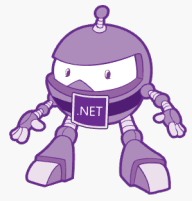
- Editor, Entry, SearchBar
  - -xf-placeholder
  - -xf-placeholder-color
- Editor and Entry
  - -xf-max-length
- NavigationPage, TabbedPage
  - -xf-bar-background-color
  - -xf-bar-text-color
- ScrollView and StackLayout
  - -xf-orientation
- ScrollView
  - -xf-horizontal-scroll-bar-visibility
  - -xf-vertical-scroll-bar-visibility
- Slider
  - -xf-min-track-color
  - -xf-max-track-color
  - -xf-thumb-color
- StackLayout
  - -xf-spacing



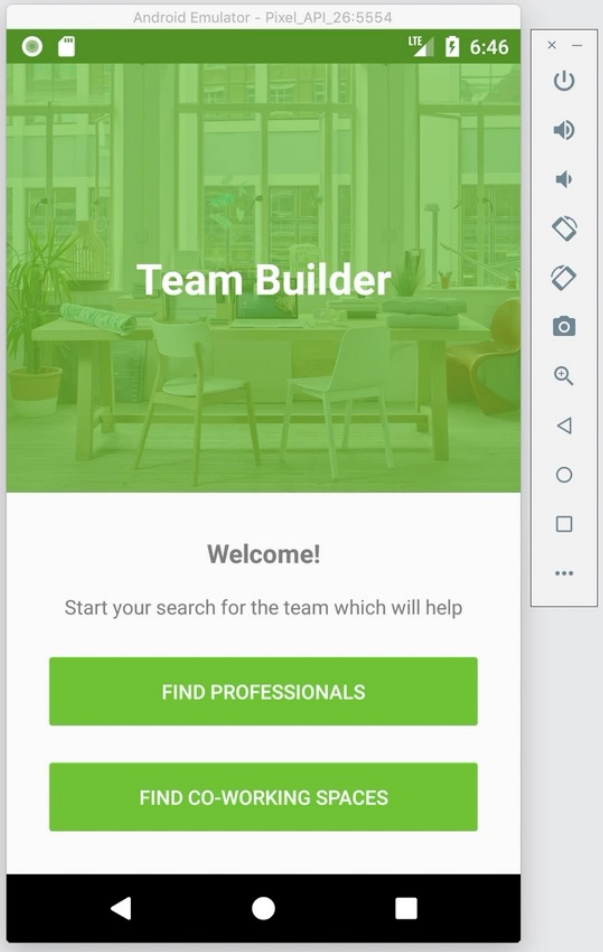
CollectionView

Perf

- ListView but better
- RecyclerView on Android
- You provide the layout
  - Grid
  - Horizontal
  - Vertical
  - FlexLayout (?)
- Context Menus
- Gestures

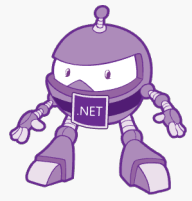


Demo – CollectionView



```
<CollectionView x:Name="Spaces" Margin="0,5,0,0”/>

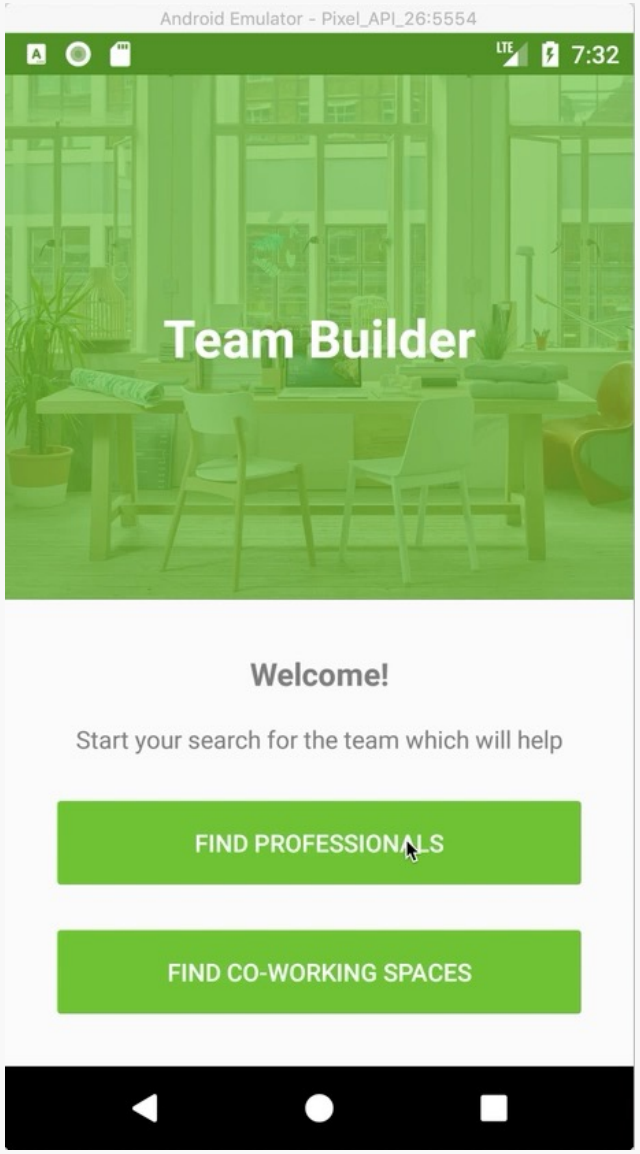
Spaces.ItemsLayout = new GridItemsLayout(2, ItemsLayoutOrientation.Vertical);
Spaces.ItemTemplate = new DataTemplate(() => new SpaceCard());
Spaces.ItemsSource = items;
```

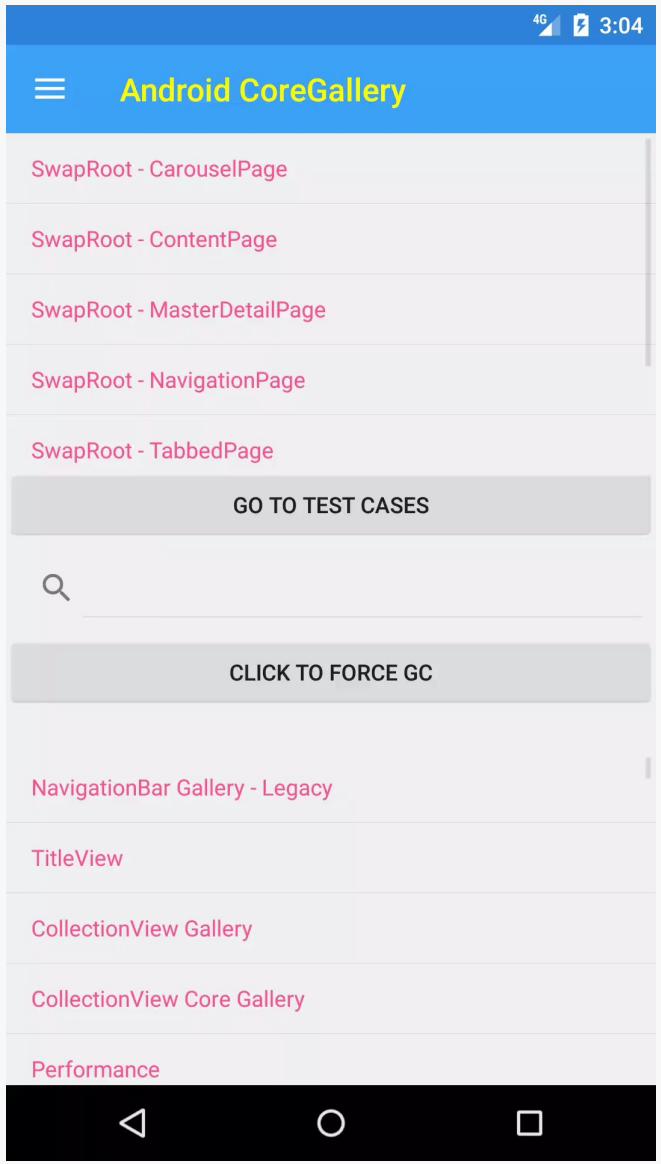
Then it dawned on us.

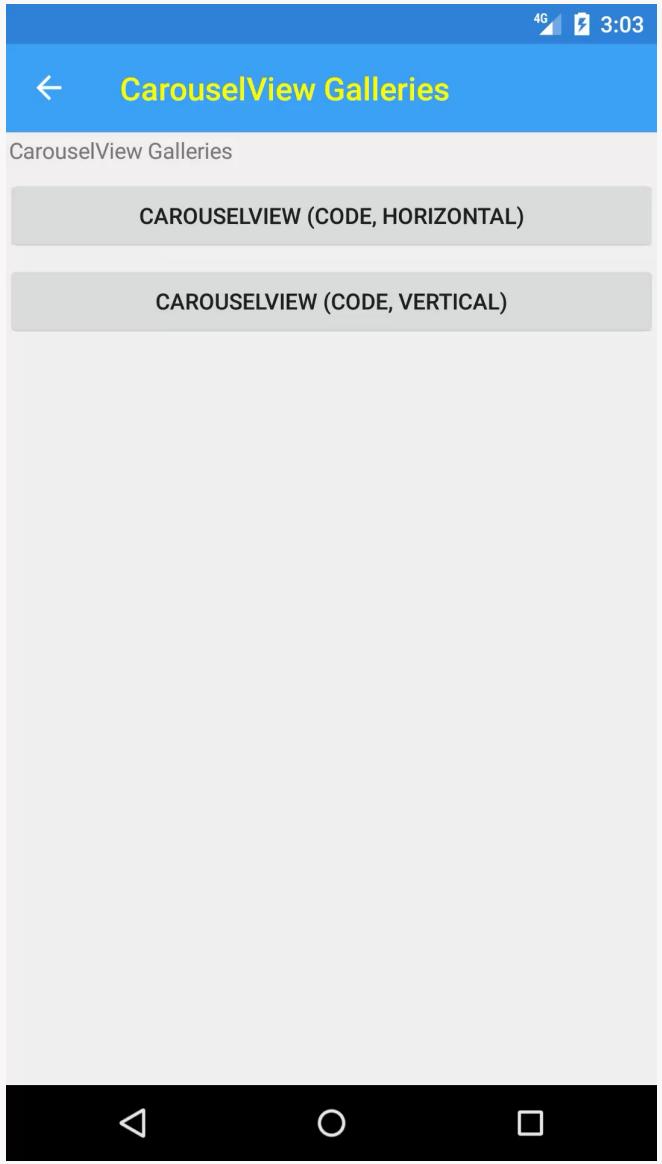
CollectionView also affords something else…

Demo – CarouselView



Demo – CarouselView

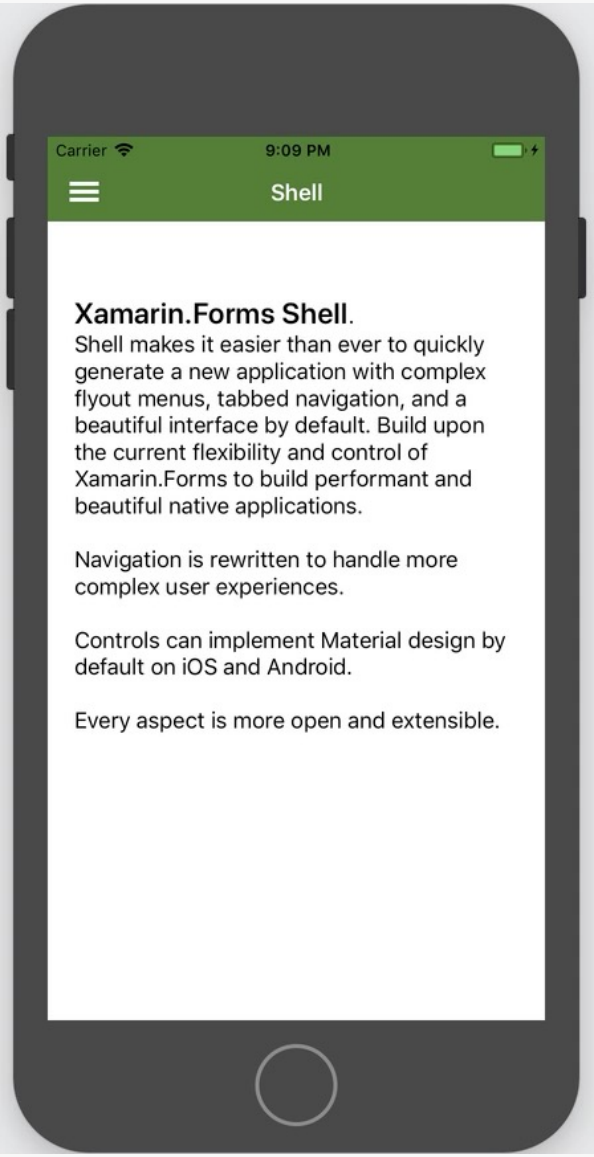




.NET Conf 2018

Xamarin.Forms Shell

Perf



- Performant by default
- Same design on iOS and Android via MaterialShell
- Platform design via Shell
- Easily describe your entire app in one file
- Flyout and Tabbed navigation
- URL routed navigation, deep linking
- Back navigation handling
- Integrated Search handler
- Snackbar
- Bottom Sheet
- Floating Action Button
- Left Bar Button
- Screen segues
- Screen transitions
- Plus everything you can already do today in Xamarin.Forms

```
<?xml version="1.0" encoding="UTF-8"?>
<Shell xmlns=…>
    <Shell.Resources>
        <ResourceDictionary Source="Styles/Global.xaml"/>
    </Shell.Resources>

    <ShellItem Route="main" Title="Home">
         <ShellSection Route="home" Title=“Home”>
            <ShellContent ContentTemplate="{DataTemplate local:MainPage}" />
        </ShellSection>
        <ShellSection Route="activity" Title="Activity">
            <ShellContent Route="shared" Title="Shared" ContentTemplate="{DataTemplate local:ActivityPage}" />
            <ShellContent Route="notifications" Title="Notifications" ContentTemplate="{DataTemplate local:NotificationsP
        </ShellSection>
        <ShellSection Route="updates" Title="Updates">
            <ShellContent ContentTemplate="{DataTemplate local:UpdatesPage}" />
        </ShellSection>
    </ShellItem>
</Shell>
```

```
<Shell.FlyoutHeader>
    <ContentView HeightRequest=“300”>
        <Label Text=“Hello XamStore” …/>
    </ContentView>
</Shell.FlyoutHeader>

<ShellItem Route="main" Title="Home">
    <ShellContent ContentTemplate=“{DataTemplate local:MainPage}” />
</ShellItem>

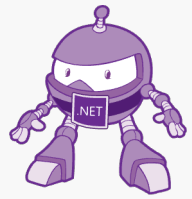
<ShellItem Route=“notifications” Title=“Notifications”>
    <ShellContent ContentTemplate=“{DataTemplate local:NotificationsPage}” />
</ShellItem>
```

Gestures

- Generic Touch
- Long Press
- Pan
  - IsPanning
- Pinch
- Rotate
- Tap
  - IsTapping
  - NumberOfTouches
  - UpdateEventArgs

Needs Your Input and Vote

https://github.com/xamarin/Xamarin.Forms/labels/a%2Fgestures



Thanks!

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@davidortinau

